

Trigonometry

MA.912.T.1 Define and use trigonometric ratios, identities or functions to solve problems.

MA.912.T.1.1

Benchmark

MA.912.T.1.1 Define trigonometric ratios for acute angles in right triangles.

Benchmark Clarifications:

Clarification 1: Instruction includes using the Pythagorean Theorem and using similar triangles to demonstrate that trigonometric ratios stay the same for similar right triangles.

Clarification 2: Within the Geometry course, instruction includes using the coordinate plane to make connections to the unit circle.

Clarification 3: Within the Geometry course, trigonometric ratios are limited to sine, cosine and tangent.

Connecting Benchmarks/Horizontal Alignment

- MA.912.GR.1.1, MA.912.GR.1.2, MA.912.GR.1.6
- MA.912.GR.7.2, MA.912.GR.7.3

Terms from the K-12 Glossary

- Angle
- Hypotenuse
- Right Triangle

Vertical Alignment

Previous Benchmarks

- MA.8.GR.1.1
- MA.8.GR.2.4

Next Benchmarks

- MA.912.T.1.5, MA.912.T.1.6, MA.912.T.1.7, MA.912.T.1.8
- MA.912.T.2
- MA.912.T.3

Purpose and Instructional Strategies

In grade 8, students solved problems involving similar triangles. In Geometry, students will use similarity in right triangles to develop the definition of trigonometric ratios. In later courses, trigonometric ratios will be used to define trigonometric functions on the unit circle and the real line.

- Instruction includes students developing the definitions for sine, cosine and tangent using ratios of side lengths in similar triangles and the Pythagorean Theorem.
 - For example, given a set of similar right triangles each with one side length missing, students can find the missing side lengths using the Pythagorean Theorem. Additionally, students can determine the constant of proportionality among the similar triangles. Students can then calculate the trigonometric ratios for each triangle and conclude that the corresponding ratios are the same for each triangle. Lead students to conclude the trigonometric ratios correspond to the angle measure and that each angle measure of an acute angle can be assigned to a sine, cosine, and tangent using an appropriate right triangle.
- Instruction includes discussing the relationships between the sine, cosine and tangent of the two acute angles of a triangle.
 - For example, given a right triangle ABC , where angle C is the right angle, students

can determine that $\sin A = \cos B$, $\cos A = \sin B$ and $\tan A = \frac{1}{\tan B}$. Students should realize that these equations are true as long as angle A and angle B are complementary angles.

- Instruction includes the use of technology, including a calculator, to show how the sine, cosine and tangent of any acute angle (and more) can be found.
- Within the Geometry course, the expectation is to use angle measures given in degrees and not in radians. Additionally, it is not the expectation for students to master the trigonometric ratios of secant, cosecant and cotangent within this course.
- It is customary to use Greek letters to represent angle measures (e.g., $\theta, \alpha, \beta, \gamma$).
- Instruction includes the connection to the unit circle.
 - For example, given a right triangle on the coordinate plane with vertices at $(0,0)$, $(x,0)$ and (x,y) , and the length of the hypotenuse equal to 1. Students can find the sine, cosine and tangent of the acute angle that the hypotenuse forms with the x -axis.

Common Misconceptions or Errors

- Students may misidentify the sides of triangles.
 - For example, students may identify the hypotenuse as being the adjacent leg or confuse the adjacent and opposite sides.

Instructional Tasks

Instructional Task 1 (MTR.2.1)

Provide students with a set of similar right triangles and their side lengths.

Part A. Identify corresponding angles and label them as $\angle A$, $\angle B$ and $\angle C$, with $\angle C$ denoting the right angle.

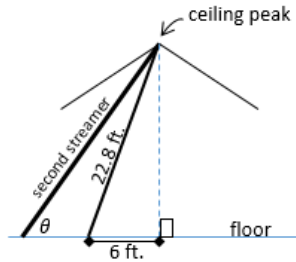
Part B. Write the following ratios for each one of the right triangles with respect to $\angle A$:
opposite leg: hypotenuse, *adjacent leg: hypotenuse* and
opposite leg: adjacent leg. What do you notice?

Part C. How do your ratios created in Part B relate to sine, cosine and tangent?

Instructional Items

Instructional Item 1

Belle is hanging streamers for her brother's surprise birthday party. She secures two streamers of different lengths at the peak of the ceiling. The center of the floor is directly underneath the ceiling peak. The distance along the floor from the center of the room to where the first streamer is attached is 6 feet. The second streamer is attached to the floor further from the center of the floor than the first streamer.

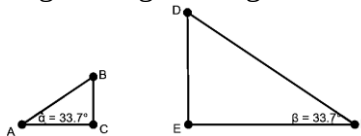


The distance between the streamers is x feet and the length of the second streamer is y feet. The angle formed between the second streamer and the floor is θ . Select all of the equations that are true to the nearest tenth based on the diagram.

- a. $\sin \theta = \frac{22.0}{y}$
- b. $\sin \theta = \frac{22.8}{y}$
- c. $\tan \theta = \frac{22.0}{6}$
- d. $\cos \theta = \frac{x}{y}$
- e. $\cos \theta = \frac{x+6}{22.8}$
- f. $\tan \theta = \frac{22.0}{x+6}$
- g. $\sin \theta = \frac{22.0}{22.8}$
- h. $\tan \theta = \frac{22.8}{x}$

Instructional Item 2

Given the diagram below showing two right triangles, complete the following statements.



Statement A. $\sin 33.7^\circ = \frac{BC}{\square}$

Statement B. $\sin 33.7^\circ = \frac{\square}{DF}$

Statement C. $\frac{BC}{AC} = \frac{\square}{\square}$

*The strategies, tasks and items included in the B1G-M are examples and should not be considered comprehensive.